# Turner Silverthorne

Website Scholar GitHub ORCID

## Contact

University of Toronto
Department of Mathematics
40 St. George Street
Toronto, Ontario, Canada

turner.silverthorne@utoronto.ca
(647)-778-8320

## Research Interests

Mechanistic models of circadian and neural systems, Circular statistics, Optimal experimental design, Scientific computing, Analysis of dynamical systems.

## Education

**University of Toronto**

Ph.D. in Mathematics — Expected May 2027
Dissertation title: *Separation of timescales in mathematical chronobiology*
Advisors: Adam Stinchcombe and Arturas Petronis

M.Sc. in Mathematics — 2022
Thesis title: *A mathematical model of promoter methylation in the circadian clock*
Advisors: Adam Stinchcombe and Arturas Petronis

**University of Waterloo**

B.Sc. in Mathematical Physics, Minor in Pure Mathematics — 2020

## Awards

| | |
|---|---|
| Queen Elizabeth II Graduate Scholarships in Science and Technology | 2026–2027 |
| NSERC Canada Graduate Scholarship | 2023–2026 |
| Blyth Fellowship, University of Toronto (declined due to NSERC) | 2023 |
| Research Excellence Award, Canadian Society for Chronobiology | 2023 |
| Ontario Graduate Scholarship, Government of Ontario | 2022 |
| Margaret Isobel Elliott Graduate Scholarship | 2022 |
| NSERC Undergraduate Student Research Award | 2019 |
| NSERC Undergraduate Student Research Award | 2017 |
| Undergraduate Research Internship, University of Waterloo | 2017 |
| Undergraduate Research Internship, University of Waterloo | 2016 |
| President's Scholarship, University of Waterloo | 2015 |

## Research Experience

| | |
|---|---|
| **Research assistant**, University of Vilnius (Life Sciences Center) | since 2022 |
| **Research trainee**, Centre for Addiction and Mental Health (Neuroscience) | since 2020 |
| **Co-op student**, Centre for Addiction and Mental Health (Neuroscience) | 2018 |
| **Co-op student**, University of Waterloo (Pure Mathematics) | 2019 |
| **Co-op student**, Institute for Quantum Computing (Pure Mathematics) | 2017 |
| **Co-op student**, Institute for Quantum Computing (Engineering) | 2016 |

## Publications

1. T Silverthorne, M Carlucci, A Petronis, AR Stinchcombe, *Optimization of experimental designs for biological rhythm discovery.* PLOS Computational Biology, **21**(11), e1013662 (2025).
2. [†]T Silverthorne and B Webster. *Gelfand-Tsetlin modules: canonicity and calculations.* Algebras and Representation Theory, **27**(2), 1405–1455 (2024).
3. [†]C Paddock, V Russo, T Silverthorne, W Slofstra. *Arkhipov's theorem, graph minors, and linear system nonlocal games.* Algebraic Combinatorics, **6**(4), 1119–1162 (2023).
4. T Silverthorne, ES Oh, AR Stinchcombe. *Promoter methylation in a mixed feedback loop circadian clock model.* Physical Review E, **105**(3), 034411 (2022).

([†]authorship in this field is alphabetical)

## Preprints

1. M Carlucci[*], ES Oh[*], T Silverthorne[*], AR Stinchcombe, M Zelnys, A Petronis. *The chronology of developing cells: are epigenomic and transcriptomic oscillations linked to their linear trajectories?.* Preprint bioRxiv:2026.04.27.720664, 2026.

([*]these authors contributed equally)

## Conferences and Talks

**Invited talks**

- [‡]Nonlinear dynamics in Biology, Canadian Mathematical Society Winter Meeting, Montreal 2026

**Contributed talks**

- Trainee Symposium, Canadian Society for Chronobiology Biennial Meeting, Montreal 2025
- Canadian Society for Chronobiology Biennial Meeting, Montreal 2025
- SRBR Biennial Meeting, San Juan, Puerto Rico 2024
- AMMCS Biennial Meeting, Wilfrid Laurier University 2023
- Canadian Society for Chronobiology Biennial Meeting, University of Guelph 2023

**Posters**

- CAIMS Annual Meeting (Virtual) 2021
- Canadian Mathematics Society Summer Meeting (Virtual) 2021
- Canadian Association of Physicists Congress, Queen's University 2017

**Other talks**

- Bird's Eye Conference 2025 & 2026
- Toronto Applied Math Learning Seminar. 2022–2026
- Math Graduate Student Seminar, University of Toronto. 2021 & 2026
- Physics of Information Lab, University of Waterloo. 2019

([†] confirmed, upcoming)

## Professional Activities

Reviewer for Frontiers in Photobiology
Reviewer for Biometrical Journal

## Committee Membership

**Bird's Eye Conference**, University of Toronto 2023–2024
*Director (2024), Co-organizer (2023)*

- Organized a two-day graduate conference for 60 attendees and 30 speakers, focusing on accessible communication between different fields of mathematics.
- Incorporated mentorship in the conference design, so that speakers would receive feedback and opportunities to improve the clarity of their presentations.
- Built a long-term framework for the event, including a mission statement and succession plan, which has allowed the conference to continue under new leadership.

**Math Department Diversity and Equity Committee**, University of Toronto. 2021–2022
*Graduate Student Representative*

- I served as a representative of the math graduate student community and contributed to a revised welcome package for graduate students emphasizing our community values, advice from senior students, and resources to promote diversity in the community.

## Outreach

*Graduate student mentor*, University of Toronto since 2023

- Mentored new graduate students in the math department.

*Math mentor*, Math Outreach Office, University of Toronto 2021

- Guided two high school students on independent study projects.

*Workshop Facilitator*, FemPhys, University of Waterloo 2019–2020

- Developed and presented three interactive workshops on tools and strategies for undergraduate research.
- Provided opportunities for students with various backgrounds to discuss and learn from one another's research experiences.

*Mentor*, Let's Talk Science, University of Waterloo 2016–2018

- Conducted weekly mentoring sessions with a gifted middle school student focused on university-level topics in mathematics and physics.
- Presented an original online physics demonstration to a highschool in Nunavut.

## TA Experience

University of Toronto

| | |
|---|---|
| Ordinary Differential Equations for Engineers (MAT 292) | Fall 2026 |
| Mathematical Modelling (APM 348) | Winter 2021 |
| Calculus I (MAT 135) | Fall 2020 |

## Graduate Coursework

**University of Toronto**

- Linear algebra and optimization
- Partial differential equations (I & II)
- Real analysis (I & II)
- Introduction to nonlinear evolution PDEs

- Numerical methods for mathematical modelling
- Cellular and molecular biophysics
- Foundational Discoveries in Genome Biology and Bioinformatics
- Deep learning: theory and data science

**University of Waterloo**
- Introduction to Lie groups and Lie algebras
- Introduction to General Relativity for Cosmology
- Applied Stochastic Processes

# References

**Adam Stinchcombe**

Associate Professor, Department of Mathematics
University of Toronto
stinch@math.toronto.edu

**Arturas Petronis**

Professor, Departments of Psychiatry and Pharmacology & Toxicology
University of Toronto

Senior Scientist and Head, Krembil Family Epigenetics Laboratory
Centre for Addiction and Mental Health (CAMH)

Professor, Vilnius University
art.petronis@camh.ca

**Andreas Hilfinger**

Associate Professor, Department of Chemical and Physical Sciences
University of Toronto
andreas.hilfinger@utoronto.ca

**Mary Pugh**

Professor, Chair, and Graduate Chair, Department of Mathematics
University of Toronto
mary.pugh@utoronto.ca